

Flask Main Application - Project Documentation (Reddy)

1. Introduction

This document describes the main application logic for the Flask project. It integrates two modular blueprints: one for the login system and another for the search engine. The system uses session-based authentication and redirects users accordingly.

2. Main Components

1. Flask App Initialization:

- Creates an instance of Flask.
- Sets a secret key for sessions.

2. Blueprints:

- login_blueprint: Handles user authentication and OTP verification.
- search_blueprint: Handles the search engine logic and database.

3. Session Handling:

- If a user is logged in, redirect to search.
- Otherwise, redirect to login page.

3. Blueprint Registration

The application uses modular blueprints for better separation of concerns.

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- ``login_blueprint``: Imported from ``login/app.py`` and registered at the root URL.
- ``init_db(app)``: Imported from ``engine/app.py``, which also registers the search blueprint.

This makes the application scalable and easier to maintain.

4. Environment Variables

The secret key for the Flask session is set via an environment variable:

- `FLASK_SECRET_KEY`: A secure random string used to sign session cookies.

You can set it in your ``.env`` file or system environment:

`FLASK_SECRET_KEY=your_secure_random_key`

5. Running the App

To run the application:

1. Set the environment variables (EMAIL credentials, `FLASK_SECRET_KEY`, API keys).
2. Ensure the database and directories are correctly structured.
3. Run the application:

```
python app.py
```

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The application will start in debug mode for development.

6. Conclusion

This main file is the entry point to the full Flask application. It controls session-based access and integrates both login and search functionality into one coherent user flow.